Remote Sensing of Environment 333 (2026) 115134

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

Nationwide mapping and characterization of land subsidence in the United States using InSAR

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ARTICLE INFO

Editor Name: Dr. Jing M. Chen

Keywords:
Land subsidence
United States
InSAR
Deep learning
Atmospheric delay

ABSTRACT

Land subsidence (LS) threatens public safety and sustainable socioeconomic development across the United States (US). Wide-area LS mapping plays a crucial role in improving the characterization of LS, thus contributing to a better understanding of its impacts beyond national borders. However, a nationwide LS map for the US is still lacking, and mapping LS at this scale faces challenges such as atmospheric artifacts, computational limitations, and data calibration issues. We develop a strategy for mapping and identifying LS on a nationwide scale. This strategy includes average deformation calculation, mosaicking of Interferometric Synthetic Aperture Radar (InSAR) results, and LS areas extraction using deep learning. Using this strategy and extensive SAR data, we present the first high-resolution, nationwide LS map for the contiguous U. S. (CONUS), derived from 23,391 ascending Sentinel-1 images acquired between 2019 and 2021. Our analysis reveals that LS affects all 48 states and the District of Columbia, covering ~45,666 km^2. Approximately 73 % of LS occurs in cultivated lands, 10 % in wetlands, with groundwater overexploitation being the predominant driver of subsidence. These subsiding areas expose 6.26 million people, 160,071 buildings, and critical infrastructure, including thousands of kilometers of railways and roads. Although some measures have been implemented to mitigate LS, existing cases suggest that long-term groundwater management is essential, and significant challenges remain in addressing severe LS. This national-scale assessment provides critical data for targeted management and informs future monitoring and mitigation strategies.

1. Introduction

Land subsidence (LS), the downward movement of the Earth's surface, is primarily caused by soil compaction or the depletion of subsurface resources, such as groundwater or hydrocarbons (Galloway and Burbey, 2011). LS results from both natural and anthropogenic processes, posing significant risks, including infrastructure damage, increased flood vulnerability, reduced land value, and threats to public health (Herrera-García et al., 2021). In the United States (US), LS has affected over 44,000 km^2 across 45 states, with annual economic losses from LS-induced flooding and structural damage exceeding $125 million (Galloway et al., 2008). As the nation faces both economic recovery and the intensifying challenges posed by climate change, the prevalence and severity of LS is anticipated to increase, potentially undermining sustainable development. Effective wide-area LS monitoring is therefore crucial to mitigate the associated risks and inform long-term resilience strategies. The availability of extensive public data in the US facilitates the characterization of LS from a macro perspective. Consequently, conducting wide-area LS monitoring within the US not only benefits the country itself but also enhances our understanding of LS in regions beyond its borders. Together, these factors make the US an ideal region for studying wide-area LS.

Interferometric Synthetic Aperture Radar (InSAR) has emerged as a critical tool for monitoring LS due to its broad spatial coverage, high

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https://doi.org/10.1016/j.rse.2025.115134
Received 10 March 2025; Received in revised form 4 October 2025; Accepted 5 November 2025
Available online 14 November 2025
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precision, and operational efficiency (Meng et al., 2024; Raspini et al., 2022). InSAR has been successfully applied at country- and continental-scales in regions such as Italy (Costantini et al., 2017), Germany (Kalia et al., 2017), Japan (Ferretti et al., 2019), Norway (Dehls et al., 2019), Netherlands (Gee et al., 2019), Greece (Papoutsis et al., 2020), Qatar (Emil et al., 2021), New Zealand (Hamling et al., 2022), Iceland (Cao et al., 2023), Iran (Haghighi and Motagh, 2024), and China (Ao et al., 2024; Zhang et al., 2024), as well as across Europe (Lanari et al., 2020). The European Ground Motion Service (EGMS), for example, uses Sentinel-1 data to monitor ground deformation across the European Union, providing valuable data for urban planning, infrastructure development, and disaster early warning (Costantini et al., 2021; Costantini et al., 2022). In the US, InSAR has been extensively used to map LS (Table 1) and identify its causes in regions such as Arizona's Willcox Basin (Conway, 2016; Peng et al., 2022; Peng et al., 2024), California's Central Valley (Carlson et al., 2024; Gualandi and Liu, 2021; Jones et al., 2022; Kang and Knight, 2023; Murray and Lohman, 2018; Neely et al., 2021; Ojha et al., 2018), Santa Clara Valley (Chaussard et al., 2014), Imperial Valleys (Jiang and Lohman, 2021), and the Los Angeles and Santa Ana Basins (Lanari et al., 2004; Riel et al., 2018; Zhang et al., 2012), as well as in Houston (Liu et al., 2022b; Qu et al., 2015), the Permian Basin (Karanam and Lu, 2023), and the Gulf Coast (Qiao et al., 2023; Qu et al., 2023). Several studies have utilized high spatial resolution deformation maps from InSAR and Global Navigation Satellite System (GNSS) to project the regions and populations that may be affected by relative sea level rise (Blackwell et al., 2020; Buzzanga et al., 2023; Creel et al., 2024; Govorcin et al., 2025; Ohenhen et al., 2023a; Ohenhen et al., 2023b; Ohenhen et al., 2024; Shirzaei and Bürgmann, 2018). Some scholars have assessed the risks of LS in major cities (Ohenhen et al., 2025). Despite these advancements, a national-scale map of LS rates across the US remains lacking.

Table 1
A summary of the local/regional LS studies in the US using InSAR.

Authors	Study sites/ regions	Main work
Conway, 2016; Peng et al., 2022; Peng et al., 2024	Willcox Basin in Arizona	Investigating the relationship between subsidence and groundwater decline. Predicting deformation time series.
Carlson et al., 2024; Gualandi and Liu, 2021; Jones et al., 2022; Kang and Knight, 2023; Murray and Lohman, 2018; Neely et al., 2021; Ojha et al., 2018	California's Central Valley	Analyzing subsidence patterns and deformation monitoring. Investigating subsidence caused by groundwater extraction.
Chaussard et al., 2014	Santa Clara Valley	Conducting hydrological and deformation analysis.
Jiang and Lohman, 2021	Imperial Valleys	Developing algorithms for InSAR data processing and deformation monitoring.
Lanari et al., 2004; Riel et al., 2018; Zhang et al., 2012	Los Angeles and Santa Ana Basins	Investigating the relationship between subsidence and groundwater decline.
Liu et al., 2022b; Qu et al., 2015	Houston	Analyzing subsidence patterns and deformation monitoring. Investigating the relationship between subsidence and groundwater decline.
Karanam and Lu, 2023	Permain Basin	Estimating deformation induced by hydrocarbon production.
Qiao et al., 2023; Qu et al., 2023	The Gulf Coast	Identifying drivers of LS.
Blackwell et al., 2020	California's coast	Evaluating the impacts of relative sea level rise.
Buzzanga et al., 2023	New York city	
Creel et al., 2024	Alaska's Arctic coastal zone	
Govorcin et al., 2025	California's coast	
Ohenhen et al., 2023b; Ohenhen et al., 2024	US coasts	
Shirzaei and Bürgmann, 2018	San Francisco Bay Area	
Ohenhen et al., 2025	28 cities	Assessing risk levels.

Developing a nationwide LS map for countries with extensive territories, such as the US, presents several significant challenges. These include the correction of atmospheric artifacts, computational limitations, and data calibration issues. As a result, the spatial distribution of LS, its underlying drivers, and the populations and infrastructure at risk remain insufficiently characterized. This study aims to fill this gap by mapping LS across the contiguous United States (CONUS) using Sentinel-1 images. This objective is achieved by developing a strategy that includes average deformation rate calculation, mosaicking of InSAR results, and efficient extraction of LS areas. We apply Multi-temporal InSAR (MT-InSAR) method to these SAR images to produce the first national-scale map of non-tectonic deformation rate for the CONUS, calculated at a resolution of 40 m. To ensure consistency in the LS reference datum, we implement a block adjustment method, validated against GNSS stations from the Nevada Geodetic Laboratory. A convolutional neural network is then employed to identify LS areas, providing the first comprehensive quantification of LS across the CONUS. Furthermore, we recognize the role of agriculture in LS, assess the exposure of populations, buildings, and roads in subsiding zones, and investigate the main potential factors associated with LS. We also examine the deformation changes in key LS regions following groundwater regulation, providing insights into the broader implications of our findings.

The remainder of this study is organized as follows: Section 2 describes the SAR images used, the wide-area InSAR data processing method, and the LS extraction method. Section 3 presents the InSAR-derived surface deformation rates for the CONUS, and analyzes national LS patterns, the critical role of cultivated land in LS, as well as the exposure of populations, buildings, and roads to LS. Section 4 discusses the usage of the deformation data, the associated risks and mitigations for LS, an evaluation of methodological effectiveness, and a comparative analysis of the results. Finally, Section 5 concludes the study.

2. SAR data and methods

2.1. SAR dataset

The used SAR data consist of ascending Sentinel-1 images acquired between January 2019 to December 2021, totaling 23,391 scenes from 28 different paths (Fig. S1 and Table S1). The interferometric process was performed using Alaska Satellite Facility (ASF) On Demand Products platform. Given the small perpendicular baselines in Sentinel-1 SAR data, we focused exclusively on the temporal baseline, setting a maximum threshold of 36 days to select interferometric pairs. The On Demand Products include unwrapped interferograms, Digital Elevation Model (DEM), amplitude files, coherence files, incidence angles, and heading angles in the UTM coordinate system. The downloaded interferograms were processed with a multi-look ratio of 10: 2 (range direction to azimuth direction), resulting in a spatial resolution of 40 m.

2.2. Wide-area InSAR data processing method

The developed InSAR data processing strategy consists of three main steps (Fig. 1). The first one involves the calculation of the average surface deformation rate. The second step is the mosaicking of InSAR deformation rates from different frames. The third step focuses on the extraction of LS areas and the time series analysis of the Area of Interest (AOI). A detailed description of the strategy is provided below.

2.2.1. Calculation of the average surface deformation rates

We employed the InSAR technique to assess surface deformation rates across the CONUS, where high-density monitoring is essential for

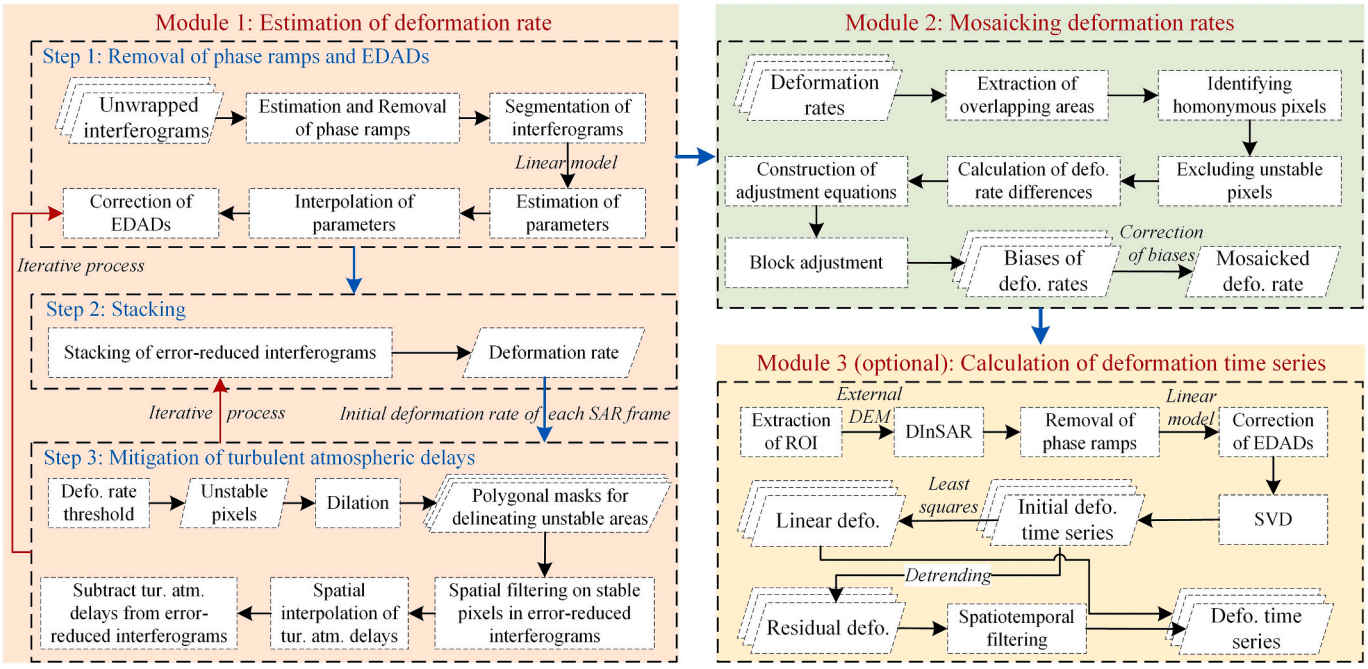

Fig. 1. Flowchart illustrating the processing of wide-area InSAR data.

identifying deformation zones and evaluating their impacts. To improve processing efficiency, we adopted the enhanced Stacking method for deformation rate estimation, followed by the Small Baseline Subset InSAR (SBAS-InSAR) method (Berardino et al., 2002) for time-series analysis in AOI. Three primary error sources in the interferograms—phase ramps, elevation-dependent atmospheric delays (EDADs), and turbulent atmospheric delays—were addressed prior to averaging the unwrapped interferograms. Phase ramps were corrected using a polynomial phase model.

2.2.1.1. EDAD correction. EDADs can be modeled as the relationship between the unwrapped phase and elevation, as expressed by the equation (Bekaert et al., 2015):

$$\varphi_{top_atm} = \varphi_0 + k(h_0 - h)^{\alpha} \tag{1}$$

where φ_{top_atm} represents the estimated EDAD, h denotes elevation of pixel, φ_0 and k are the unknown parameters φ_0 is the phase delay at h_0. Since this model is specifically designed for EDAD correction, applying it to pixels that do not exhibit EDADs or are affected by significant interfering signals, such as turbulent atmospheric delays or deformations, may lead to inaccurate corrections or the unintended removal of deformation signals. Specifically, our primary goal is to extract LS, many of which occur in flat areas where EDADs are usually negligible. To prevent the mis-removal of deformation signals or the introduction of errors, we propose an adaptive EDAD correction method that only corrects EDADs in affected pixels. The details of this method are described as follows.

Since EDADs cannot be corrected using a fixed model when the interferogram is large (Liang et al., 2019). To correct EDADs, we divided the interferogram into multiple 6 km × 6 km patches, with adjacent patches have 50 % overlapping rate. This patch size was chosen based on prior research, which suggested that this size minimizes the root mean square error (RMSE) after EDAD correction (Wang et al., 2022a). Fitting between the unwrapped phase and elevation was performed within each patch. For each patch, the parameters k and φ_0 were initially estimated using the Iteratively Reweighted Least Squares (IRLS) method. The performance of the fitted model was then evaluated using the quality metric Q, defined as:

$$Q = \sigma_{top_atm} / (k\sigma_h) \tag{2}$$

where Q represents the quality of the fitted model, σ_{top_atm} is the standard deviation (SD) of the residuals between the modeled EDADs and the observed unwrapped phases, and σ_h is the SD of elevation. In Eq. (2), σ_{top_atm} accounts for unmodeled phase components, such as turbulent atmospheric delays. A larger σ_{top_atm} indicates poorer model performance. $k\sigma_h$ reflects the magnitude of EDAD variations attributable to elevation changes. Thus, a lower Q indicates better model performance. Patches exhibiting poor fitting performance, defined as those with Q greater than a set threshold (10 in this study), were identified. For such patches, the parameters k and φ_0 were set to zero. Following parameter estimation, the parameters derived for each patch were assigned to its central pixel and spatially interpolated across the remaining pixels. Finally, the EDAD of each pixel was estimated using Eq. (1) and subsequently removed from the unwrapped phase.

2.2.1.2. Estimation of average deformation rates. After correcting for phase ramps and EDADs, the main remaining phase components in the interferograms consist of turbulent atmospheric delays, deformation signals, and noise. Since only deformation exhibits temporal correlation, interferogram stacking can be employed to suppress the influence of atmospheric artifacts and random noise. The stacking process aims to solve the least squares minimization problem given be Eq. (3):

$$\min \sum_{i=1}^{N} \left(\varphi_i - \frac{4\pi}{\lambda} V_{LOS} \Delta t_i \right)^2 \tag{3}$$

where N is the number of interferograms, φ_i represents the unwrapped phase of the i-th interferogram, λ is the radar wavelength, V_{LOS} is the desired deformation rate in the Line of Sight (LOS) direction, and Δt_i is the temporal baseline of the i-th interferogram. By solving Eq. (3), we derive Eq. (4) to estimate the average deformation rate for each pixel.

$$V_{LOS} = \frac{\sum_{i=1}^{N} w_i \Delta t_i \varphi_i}{\sum_{i=1}^{N} w_i \Delta t_i^2} \tag{4}$$

where w_i represents the weights that are determined by coherence.

2.2.1.3. Iterative process and mitigation of turbulent atmospheric delays. When correcting for EDADs, the unwrapped phase of pixels may include additional components, such as deformation signals, which could introduce biases into the estimated parameters. Given the lack of prior knowledge regarding deformation areas across the entire US, we implemented an iterative estimation approach for EDAD correction. The procedure consisted of the following steps: First, we applied the aforementioned method to estimate and remove the initial EDAD components. The corrected interferograms were then used to compute preliminary deformation rates using Eq. (4). These initial deformation rates served as prior information to identify potential deformation zones, allowing us to mask pixels exhibiting significant deformation signals. To systematically identify moving pixels, we established an empirical threshold based on the SD of the initial deformation rate estimates, masking pixels that exhibit absolute deformation rates greater than the SD. Additionally, to ensure comprehensive coverage of deformation areas, we performed morphological dilation on the masked pixels to obtain complete deformation boundaries. After the masking process, we repeated the phase ramp and EDAD estimation using only the non-deforming pixels. The final EDADs were then subtracted from each pixel's unwrapped phase.

Following EDAD correction, the interferograms were reprocessed to estimate deformation rates for each frame using Eq. (4). Once the updated deformation rates were obtained, moving pixels were identified and masked according to the previously described methodology. To isolate turbulent atmospheric components, we applied a spatial low-pass filtering operation to the remaining stable pixels. The filtered output was then spatially interpolated across all pixels to reconstruct complete atmospheric delay fields. These estimated turbulent atmospheric phases were subtracted from the EDAD-corrected interferograms to effectively reduce atmospheric artifacts. The final processing step involved computing precise deformation rates from the atmospherically corrected interferograms by reapplying Eq. (4).

2.2.2. Calibration and mosaicking of InSAR deformation rate data

We processed InSAR data from 321 frames, with deformation results in each frame aligned to a reference point within the frame. However, variations in reference points across frames can introduce inconsistencies in deformation rate estimates. To address this issue, block adjustment (Wang et al., 2024; Wang et al., 2022b) was applied, which reduces deformation rate discrepancies between frames.

Under ideal conditions, the deformation rate of the same ground point, assuming no deformation, should be identical across overlapping frames. Thus, homonymous points in the overlapping regions of different frames satisfy the relationship expressed in Eq. (5):

$$v_{i,k} - v_{j,k} = \delta_{i,k} - \delta_{j,k} \tag{5}$$

where $v_{i,k}$ and $v_{j,k}$ represent the deformation rates of point k in frames i and j, respectively, while $\delta_{i,k}$ and $\delta_{j,k}$ denote the deformation rates biases at point k in frames i and j, respectively. These biases primarily arise from discrepancies in reference points and can be assumed to be constant.

To formulate this as a system of Equations, we express the relationship in matrix form as:

$$\begin{bmatrix} \Delta v_1 \\ \Delta v_2 \\ \vdots \\ \Delta v_M \end{bmatrix} = \boldsymbol{B}_{M\times N} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \vdots \\ \delta_N \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_M \end{bmatrix} \tag{6}$$

or equivalently,

$$\Delta \boldsymbol{V}_{M\times 1} = \boldsymbol{B}_{M\times N}\boldsymbol{\delta}_{N\times 1} + \boldsymbol{\varepsilon}_{M\times 1} \tag{7}$$

where $\Delta \boldsymbol{V} = [\Delta v_1 \Delta v_2 \cdots \Delta v_M]^T$ represents the differences in deformation rates across frames; M is the total number of overlapping areas; N is the number of frames; $\boldsymbol{B}$ is the design matrix, with each row corresponding to an overlapping area and each column corresponding a frame; $\boldsymbol{\delta} = [\delta_1\ \delta_2 \cdots \delta_N]^T$ denotes the deformation rate biases for the N frames, $\boldsymbol{\varepsilon} = [\varepsilon_1\ \varepsilon_2 \cdots \varepsilon_M]^T$ represents the residual errors.

Considering the variation of incidence angles across frames, we only selected pixels with an absolute deformation rate less than 1 cm/y to estimate the biases. We applied a least squares method to solve Eq. (7) and estimate the deformation rate biases. The difference in deformation rates between two overlapping frames was calculated as the average of the differences across all homonymous pixels. After obtaining the biases, we adjusted the deformation rates by subtracting the corresponding bias from each pixel's deformation rate, resulting in a corrected and consistent deformation rate dataset for subsequent analysis. It is worth noting that all the InSAR deformation results in this study are in the radar LOS direction.

2.2.3. Deep learning for LS detection using InSAR data

LS is widespread across the CONUS, but the deformation field derived from InSAR data is often contaminated by spurious signals, such as atmospheric delays and noise. To address this, we developed DeforNet (Zhao et al., 2022), a deep learning model based on U-Net architecture (Akhtarmanesh et al., 2024; Ronneberger et al., 2015; Safari et al., 2024; Vafaeinejad et al., 2025), which integrates attention mechanisms and depth-wise separable convolutions for noise reduction and model efficiency.

To optimize DeforNet's deformation detection capability, we trained the model using InSAR-derived deformation rate maps spanning several Chinese provinces: Sichuan, Guizhou, Ningxia, Qinghai, Gansu, Hebei, Shanxi, and Guangdong. These training datasets encompass diverse geological setting and deformation patterns, enhancing the model's generalization capacity. The complete details regarding the training deformation areas are available through the InSAR Club repository (website: http://47.119.124.134/insarclub/). The training deformation areas were cropped into 224 × 224 pixel samples, and augmented (mirroring and rotation) to expand the dataset to 20,280 samples. Then, the dataset was split, with 80 % for training and 20 % for testing. Training was conducted over 40 epochs with a batch size of 8 and an initial learning rate of 0.0005, which was halved every 10 epochs using the Adam optimizer. The model achieved an F1-score of 0.96, demonstrating high accuracy. Since the wrapped gray images representing the rates of deformation areas derived from InSAR data exhibit strong similarities across different regions, the deep learning model trained on LS samples from other countries can be effectively transferred to identify deformation areas in the InSAR deformation map of the US.

For the InSAR data from the US, the model processed the deformation fields state-by-state. Initially, we denoised the deformation field using spatial low-pass filtering. The deformation rates were then wrapped and converted to grayscale, with pixel values normalized to the range [0, 255]. The wrapped deformation rates were divided into patches with a 25 % overlap. In the wrapped deformation filed, deformation signals appear as high-frequency signals, while noise manifests as scattered low-frequency signals. These patches were subsequently input into DeforNet for deformation area identification. The initial identification results exhibited small gaps and edge serrations. Finally, after applying expansion and smoothing processes, we reassembled the patches into the original deformation rate for each state, yielding the final identification results.

Identified deformation areas included LS, landslides, and earthquakes. Non-LS areas were excluded using Copernicus DEM to filter out regions with slopes greater than 5°, and the U. S. Geology Survey (USGS) earthquake catalog was used to exclude seismic-related deformation.

2.2.4. Data management and processing workflow

Processing over 20 thousand Sentinel-1 scenes covering the CONUS presented substantial computational challenges, requiring a systematic approach to data management and distributed processing. The workflow comprised three key phases: (1) data acquisition through the ASF On Demand platform (total ~ 87 TB raw data across 321 frames, each ~270 GB), (2) development of customized processing algorithms, and (3) implementation of high-performance computing. To manage this massive dataset, we employed multiple ASF accounts and physical storage arrays, with several researchers coordinating the download of interferograms and ancillary files. The processed data requirements exceeded 1.2 TB per frame, necessitating periodic purging of intermediate files to conserve storage space.

The processing pipeline was implemented on eight Ubuntu 20.04 workstations (each with 32-core CPUs, 64 GB RAM each) using custom MATLAB/GAMMA software. The automated workflow included: (1) format conversion from TIFF (the initial format of the downloaded files) to GAMMA-compatible files, (2) pixel quality control via coherence/amplitude dispersion thresholds, (3) error correction for phase ramps and atmospheric delays, (4) deformation rate calculation through Stacking, and (5) time-series analysis using SBAS. Final outputs were transformed from UTM (the original coordinate system of the downloaded files) to WGS84 coordinates following rigorous validation. This distributed processing strategy enabled efficient handling of the continental-scale dataset while maintaining computational tractability.

3. Results and analysis

3.1. InSAR-derived surface deformation rate of the CONUS

Fig. 2(a) and (b) show the average deformation rate of the CONUS before and after block adjustment. As seen in Fig. 2, the deformation rate after block adjustment is more consistent than that before the adjustment (detailed analysis is provided in Section 4.4). Most areas in the CONUS exhibit a stable state, with some large-scale deformation occurring in the western part of the CONUS. We used GNSS data provided by the Nevada Geodetic Laboratory (http://geodesy.unr.edu/index.php) to validate the accuracy of the InSAR results. There are 5721 GNSS stations located within the CONUS. After excluding GNSS data that didn't meet the validation criteria (Text S1), 4242 GNSS stations were retained for validation (Fig. 3). Statistical analysis reveals that 87.5 % of the GNSS stations exhibit deformation rate differences within 8 mm/y, with only 6.8 % exceeding 1 cm/y (Fig. 3(c)). The RMSE between the InSAR and GNSS deformation rates is 5.2 mm/y, confirming the reliability of the InSAR results.

3.2. National patterns of LS

This study presents the first comprehensive nationwide assessment of LS across the CONUS, highlighting its spatial distribution and characteristics. The subsidence map (Fig. 4(a)) shows that LS affects all 48

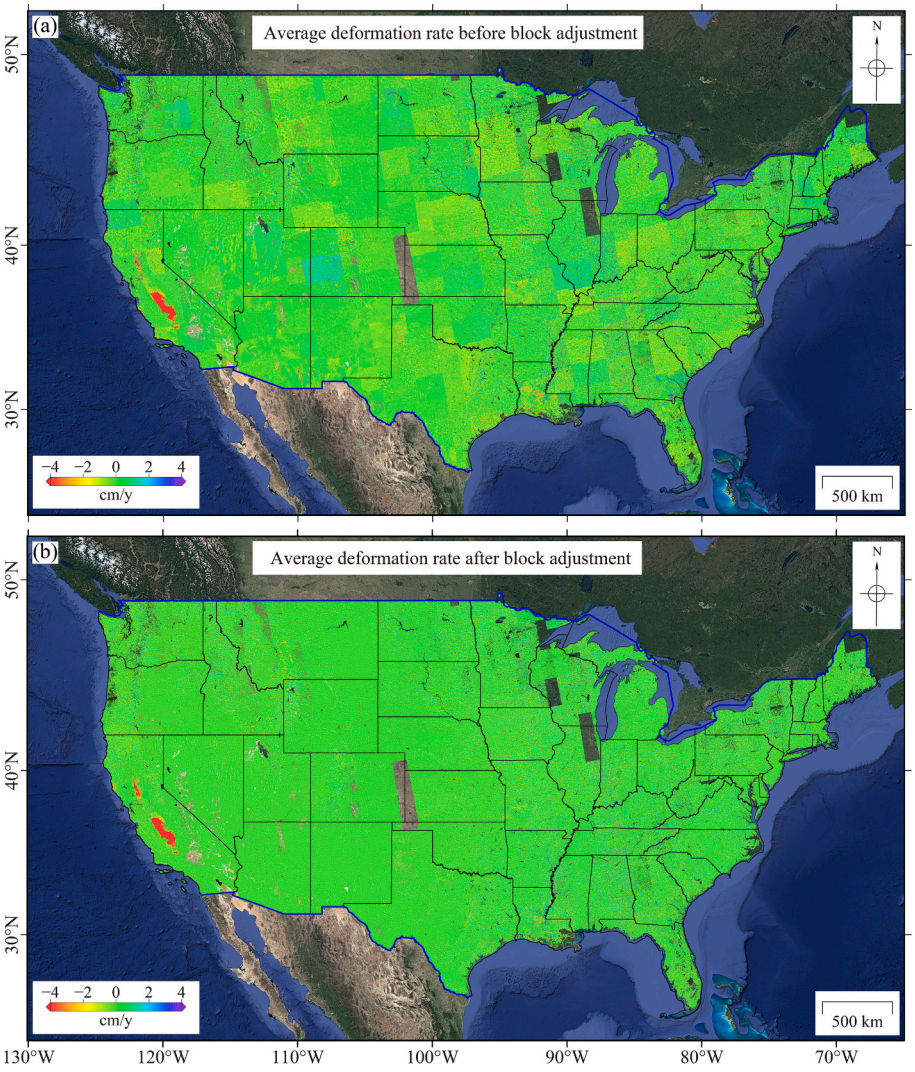

Fig. 2. Average deformation rate derived from ascending Sentinel-1 images. Deformation rate before (a) and after (b) block adjustment.

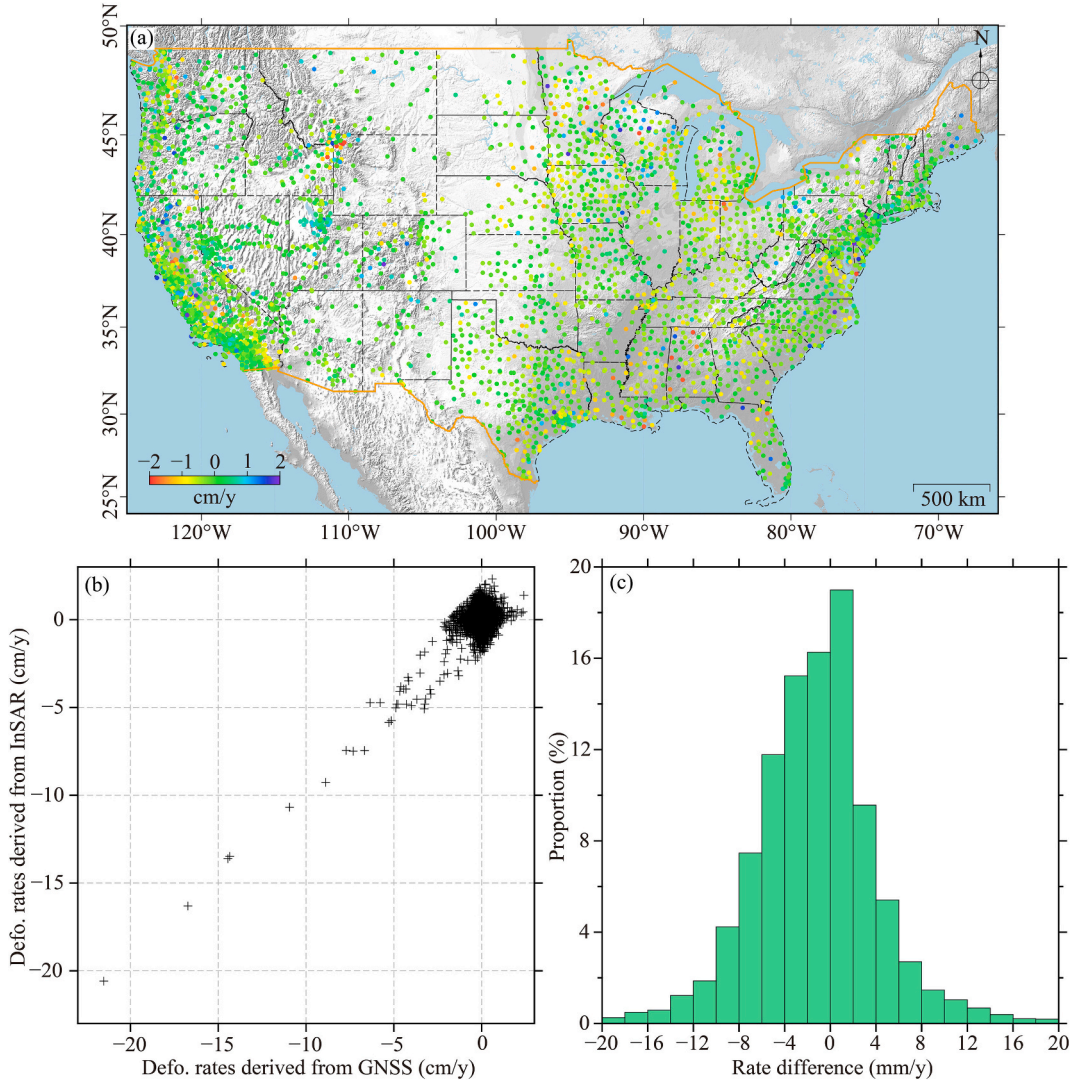

Fig. 3. Comparisons of deformation velocities derived from GNSS and InSAR. (a) Spatial distribution of differences in deformation velocities between GNSS and InSAR, with each dot representing a GNSS station. (b) Scatter plot comparing deformation rates obtained from GNSS and InSAR. (c) Histogram of deformation rate differences.

contiguous states and the District of Columbia. The mean and median deformation rates for the LS areas are −4.3 cm/y and −2.7 cm/y, respectively. Approximately 10.2 % of the LS areas exhibit deformation rates between −1 cm/y and 0 cm/y, while 20.5 % and 21.1 % experience rates between −2 cm/y and −1 cm/y, and between −3 cm/y and −2 cm/y, respectively (Fig. 5). The total area of LS is approximately 45,666 km^2, with 62.5 % located in California (Fig. 4(b)). Larger LS zones are predominantly found in the West, Midwest, and South, with California, Arizona, Florida, Louisiana, and Texas being the most affected states. In contrast, northeastern states such as Vermont, New Hampshire, and West Virginia have smaller subsidence areas (Fig. 4(b)). The identified LS areas likely underestimate the actual extent of LS due to several factors: (1) Some regions, such as the Los Angeles and Santa Ana Basins, were inactive during the InSAR monitoring period but experienced significant subsidence in the past (De Luca et al., 2017; Lanari et al., 2004; Riel et al., 2018; Zhang et al., 2012). These areas are not captured in our current analysis. (2) Some LS areas are too small to be detected by our InSAR results, or they consist of only a few pixels, making it difficult to distinguish them from background noise. Consequently, these regions are not classified as LS areas. (3) Some LS areas exhibit deformation rates that are too small to be reliably identified during the classification process. (4) Mining activities, which often occur in mountainous regions with steep slopes, can induce localized subsidence. However, due to complex topography, some mining-induced subsidence areas are not classified as LS (see Section 2.2.3).

The severity of LS varies across the CONUS. The maximum deformation rate is recorded in California's Central Valley, with a maximum subsidence rate along the LOS direction of 47.5 cm/y, corresponding to a vertical subsidence rate of approximately 58 cm/y. Quartile analysis of deformation rates in the LOS direction shows that 25 % of affected areas experience subsidence rates less than 1.6 cm/y, 50 % have rates below 2.7 cm/y, and 75 % fall below 5.4 cm/y. California, with the highest LS rates, shows that 24 % of affected areas have subsidence rates between 2.7 cm/y and 5.4 cm/y, while 38 % exceed 5.4 cm/y (Fig. 4(a)). Other states such as Oregon, Minnesota, Louisiana, Virginia, and Texas also experience significant rates of LS (Fig. 4(a)). Northeastern states like New Hampshire, Maryland, Connecticut, New Jersey, and Delaware exhibit minimal LS rates, with most affected areas experiencing subsidence rates below 2.7 cm/y (Fig. 4(a)).

LS is closely tied to land cover types (A detailed explanation of the land cover type is provided in Text S2). Of the total LS area, 73 % occurs in cultivated lands, 10 % in wetlands, 5 % in grasslands, and 4 % in artificial surfaces (Fig. 4(c)). Shrubs and forests each account for 3 %, with the remaining subsidence areas including water bodies, bare land,

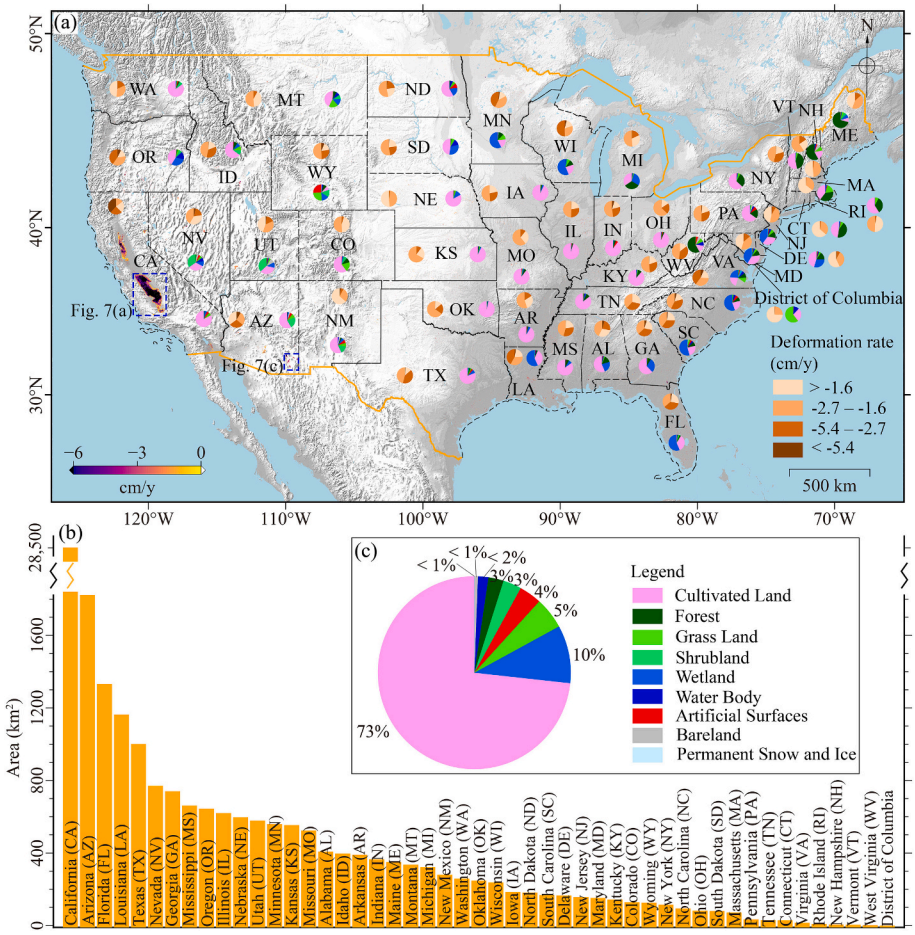

Fig. 4. Overview of LS in the CONUS. (a) LS map of the CONUS. The pie charts, each divided into four colored segments, illustrate the distribution of LS rates across the states. An additional type of pie chart, presented in a distinct color scheme, shows the proportion of various land cover types for each state, with each color corresponding to the categories represented in (c). The two blue-dotted rectangles denote the locations of LS areas in Fig. 7. (b) LS area of each state. (c) Proportion of various land cover types within LS areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

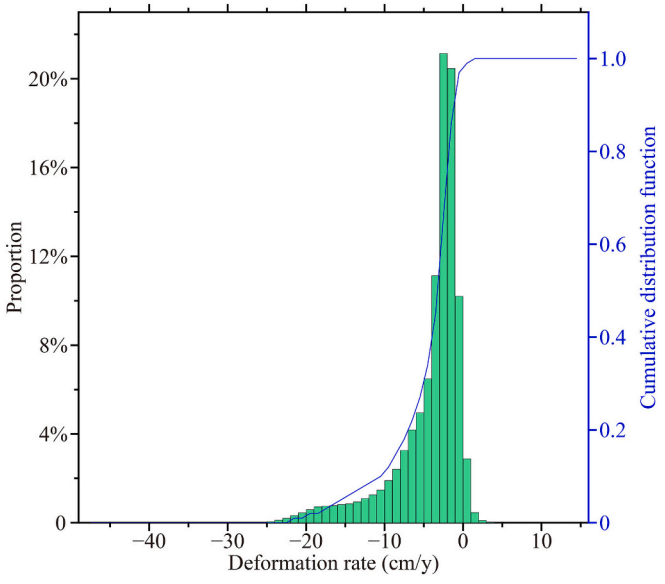

Fig. 5. Distribution of deformation rates in LS areas.

and permanent snow and ice (Fig. 4(c)). Cultivated lands account for the largest proportion of LS in 33 states, with Arkansas, Illinois, Iowa, Kansas, Ohio, and Oklahoma experiencing over 90 % of their LS occurring in agricultural areas (Fig. 4(a)). In contrast, states primarily along the East Coast and Gulf of Mexico, such as Florida, Louisiana, Maryland, Minnesota, New Jersey, North Carolina, South Carolina, Virginia, and Wisconsin, show the highest proportion of LS in wetlands (Fig. 4(a)). Subsidence in artificial surfaces is primarily caused by human activities, such as mining, engineering construction, and oil and gas extraction. The distribution of subsidence in artificial surfaces is uneven, with Wyoming being the only state where subsidence in artificial surfaces is predominant (Fig. 4(a)). In northeastern states, such as Maine, New Hampshire, and Virginia, which experience milder LS, exhibit more prominent subsidence in forested areas (Fig. 4(a)). Other northeastern states, including New York, Maryland, and Rhode Island, also show considerable subsidence in forested areas.

3.3. The critical role of cultivated land in LS

Cultivated lands are a key contributor to LS in the CONUS, accounting for approximately 73 % of the total LS area (Fig. 4(c)). A more detailed examination of subsidence in croplands provides valuable insights into the broader dynamics of LS in the region. Cropland areas (An introduction of the cropland data is provided in Text S3) are largely concentrated in the Midwest, encompassing states such as North Dakota, Iowa, Kansas, Minnesota, Illinois, Nebraska, and South Dakota,

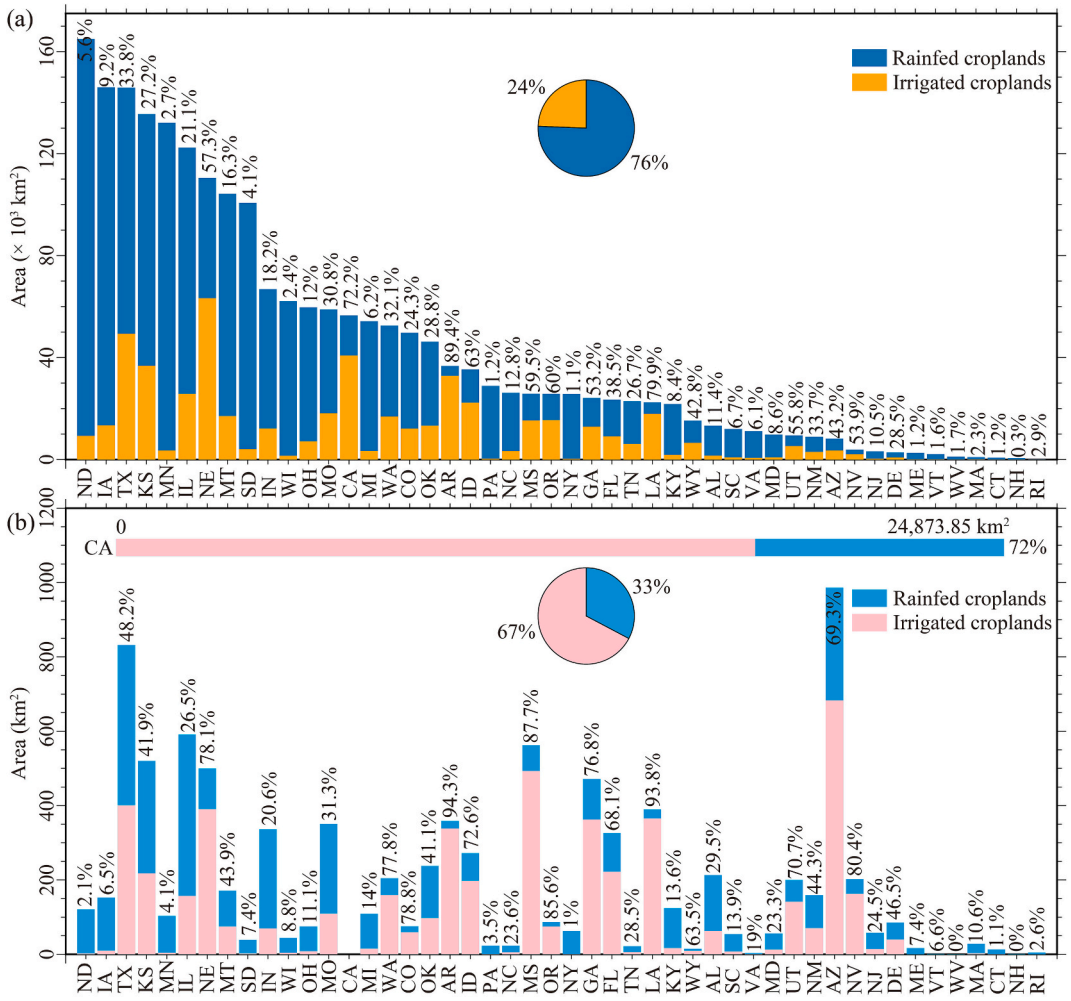

Fig. 6. Distribution of cropland types and LS in croplands across the CONUS. (a) Proportion of irrigated and rainfed croplands by state. Histogram bars show the percentage of irrigated croplands relative to total cropland area for each state; the pie chart summarizes the overall distribution of irrigated and rainfed croplands across the CONUS. (b) Proportion of LS in irrigated and rainfed croplands by state. Histogram bars show the percentage of LS in irrigated croplands relative to total LS area in croplands for each state; the pie chart summarizes the overall distribution of LS in irrigated and rainfed croplands across the CONUS.

alongside Texas in the southwest and Montana in the west (Fig. 6(a)). In contrast, states with limited cropland, predominantly in the Northeast—such as Rhode Island, New Hampshire, Connecticut, Maryland, West Virginia, Vermont, Maine, Delaware, and New Jersey—experience minor subsidence. Rainfed croplands dominate in most states, with only ten states reporting a greater extent of irrigated croplands than rainfed croplands. Interestingly, there is no straightforward correlation between the total area of croplands and the extent of LS in these regions. However, states with a higher proportion of irrigated croplands, such as California, Mississippi, and Arizona, demonstrate significantly larger subsidence areas compared to states with smaller proportions of irrigated croplands, despite ranking lower in total cropland area (Fig. 6). While irrigated croplands account for only 24 % of the total cropland area, they contribute to 67 % of the LS occurring in cultivated lands, underscoring the pivotal role of irrigation in driving subsidence across the CONUS.

Groundwater depletion, driven by excessive irrigation, is a primary cause of LS in agricultural regions. In cultivated lands experiencing LS due to groundwater overexploitation, the impacts often extend beyond the agricultural lands themselves, affecting adjacent environments such as artificial surfaces and grasslands. Our analysis of key LS regions (Fig. 7) shows that, although 73 % of LS occurs in cultivated lands, the broader impact of agricultural practices on LS is substantial. The regions depicted in Fig. 7 are largely shaped by groundwater overexploitation for irrigation (Conway, 2016; Murray and Lohman, 2018; Ojha et al., 2018; Peng et al., 2022). In California's Central Valley, two major LS zones, CA_LS1 and CA_LS2, are predominantly composed of cultivated lands. CA_LS1, the largest LS area in the CONUS, is predominantly covered by croplands (84.56 %), with grasslands, artificial surfaces, and wetlands accounting for the remaining 15.44 %. Although oil and gas extraction also contribute to subsidence in this region (Hu et al., 2021), groundwater depletion for agriculture remains the dominant factor. In Arizona (Fig. 7(c) and (d)), AZ_LS1 and AZ_LS4 exhibit significantly larger proportions of cultivated lands compared to other land cover types. In contrast, the remaining four LS zones do not show a markedly higher proportion of croplands, and in AZ_LS5, the proportion of croplands is even smaller than that of shrubs. Nevertheless, the total LS area resulting from groundwater extraction in agricultural lands is considerably larger than the cropland area itself. This analysis highlights the significant role of groundwater overexploitation in driving LS across the CONUS.

3.4. Population, buildings and roads exposed to LS

LS poses substantial risks to human populations, with approximately 6.26 million individuals residing in affected areas. California is the most affected state, representing 96 % of the exposed population (An introduction of the population data is provided in Text S3), followed by

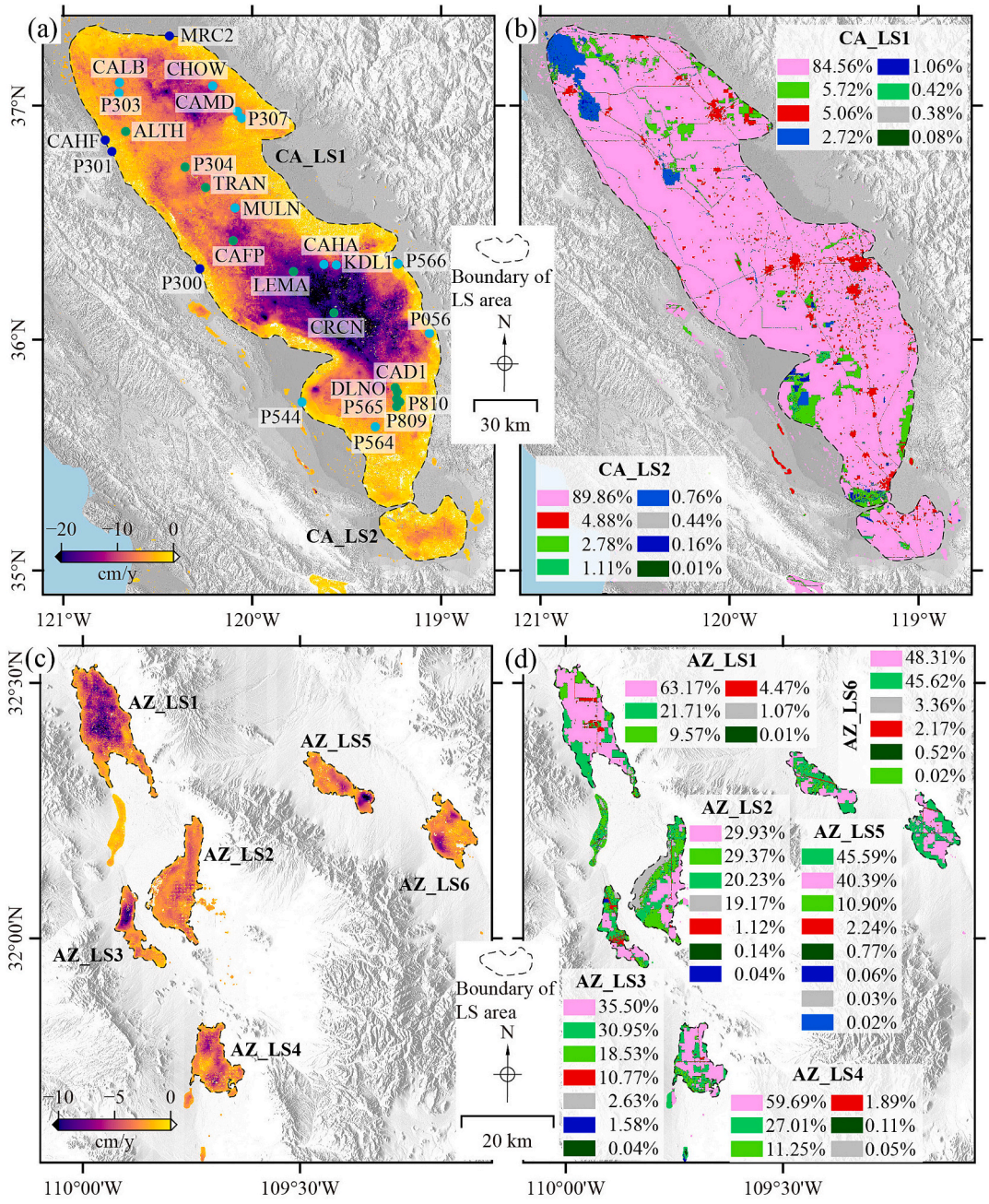

Fig. 7. Deformation rates and land cover types in LS areas driven by groundwater overexploitation. (a) Average deformation rates in two representative LS areas in California's Central Valley. Colored dots denote GNSS stations: blue (no deformation), dark green (reduced subsidence), and cyan (long-term subsidence). (b) Land cover types in the two California LS areas, with color labels consistent with Fig. 4(c). (c) Average deformation rates in six representative LS areas in Willcox, Arizona. (d) Land cover types in the six Arizona LS areas, with color labels consistent with Fig. 4(c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Arizona (87,099) and Louisiana (28,429). In total, 31 states report populations exceeding 1000 individuals exposed to LS (Table S2). To evaluate LS risks, we analyzed both differential settlement and population exposure (see Text S4). As shown in Figs. 8 and S2, regions with very low to low LS risk encompass 72.9 % of subsiding areas, while high to very high-risk zones account for only 1.6 %. Despite their limited geographic extent, these high-risk areas impact 9.6 % of the exposed population. Specifically, approximately 278,769 individuals reside in very high-risk zones, and 321,266 inhabit in high-risk areas. The majority, about 5.4 million people, are located in medium-risk zones. California dominates the high-risk landscape, encompassing 78 % of high-risk areas and 83 % of very high-risk zones. Correspondingly, 90 % of individuals residing in high-risk LS areas and 94 % of those in very high-risk regions live in California. Colorado, District of Columbia, Nevada, and West Virginia have no very high-risk LS zones. Table S2 provides a comprehensive breakdown of LS-affected areas and populations by risk level, highlighting the disproportionate exposure in California.

The angular distortions were calculated based on the differential settlement observed during the study period (~3 years). As such, these findings do not account for the total distortion in the affected areas. Given that some LS zones have experienced subsidence over several decades, the cumulative distortions could be dozens of times greater than those recorded during the study period. Therefore, the actual LS

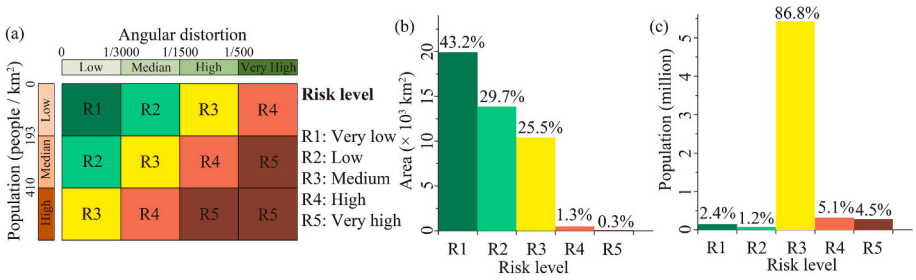

Fig. 8. LS risk levels and population exposure. (a) Risk assessment matrix defining LS risk levels based on differential settlement and population density. (b) Proportional distribution of LS-affected areas across risk levels. (c) Population exposure to LS hazards, stratified by risk level.

risk levels may be more severe than those derived from the current analysis.

Beyond its impact on populations, LS also poses significant threats to critical infrastructure, including buildings and road networks, potentially leading to substantial economic losses. Using OpenStreetMap data, we estimate that approximately 160,071 buildings are situated in LS-affected areas, of which 150,458 (94 %) are in California. Arizona and Massachusetts rank second and third, with 2832 and 735 buildings at risk, respectively. Notably, states such as Kentucky, South Dakota, and West Virginia report no buildings exposed to LS hazards. Similarly, the vulnerability of road infrastructure to LS is a cause for concern. LS-affected roads span various categories: railways (1753 km), motorways (1779 km), primary roads (1425 km), secondary roads (2766 km), and trunk roads (692 km). California dominates this exposure, with 1505 km of railways, 1675 km of motorways, 1143 km of primary roads, 2346 km of secondary roads, and 611 km of trunk roads at risk. In contrast, New Hampshire, Virginia, and West Virginia, as well as the District of Columbia, report no LS-affected roads. Additional information on threatened infrastructure, including road lengths by category, is provided in Table S3.

4. Discussion

4.1. Examples of deformation data utilization

Since the LS rate map is fundamental for hazard assessment and mitigation, the lack of on-site investigation and geological knowledge makes it difficult to perform subsequent interpretation at local scales. Therefore, we have made this dataset publicly available on our website (http://47.119.124.134/insarclub/) and encourage experts in related fields to utilize it, thereby maximizing the utility of the derived deformation rate map. The method for obtaining the data is described in Text S5. To demonstrate the application of the LS data, we present several case studies that utilize the available dataset to analyze LS from a macro perspective.

Spatial analysis shows that 91 % of LS areas are located within aquifer zones (Fig. S3). Additionally, an analysis of 48,341 groundwater wells, covering the time span of our InSAR observations, reveals that LS-affected areas exhibit significantly higher rates of surface-to-groundwater depth change (median: 11 cm/y; mean: 24 cm/y) compared to non-LS areas (median: 2 cm/y; mean: 7 cm/y) (Fig. 9 and Text S3). This case indicates a potential relationship between LS and groundwater extraction.

By overlaying oil and gas fields with the LS map of California (Fig. 10), we observe that many LS zones coincide with these fields. However, since many oil and gas fields are situated within cultivated lands, particularly in California's Central Valley, distinguishing the independent contributions of oil and gas extraction versus groundwater withdrawal remains challenging. Nevertheless, LS observed in the South Belridge oil field, primarily in artificial surface (Fig. 10), suggestions a possible correlation between hydrocarbon extraction and LS, as this oil field remains active.

We emphasize that the case studies presented above are intended solely to demonstrate the application of the LS dataset and do not establish definitive causal mechanisms. More detailed on-site investigations and models are required to confirm the drivers of LS.

4.2. Mitigation of groundwater extraction-induced LS and associated challenges

Groundwater overexploitation is a key driver of LS in the CONUS, necessitating urgent attention. Groundwater management practices vary widely across states, with California—experiencing the most severe LS—being particularly affected by recurrent droughts. In 2014, California enacted the Sustainable Groundwater Management Act (SGMA), the first state legislation to regulate groundwater extraction comprehensively (Bostic et al., 2023; Larson, 2024). SGMA encompasses a range of recharge projects, some of which have been completed and others remain underway (Larson, 2024). Despite these efforts, mitigating the long-term impacts of groundwater over-extraction on LS remains

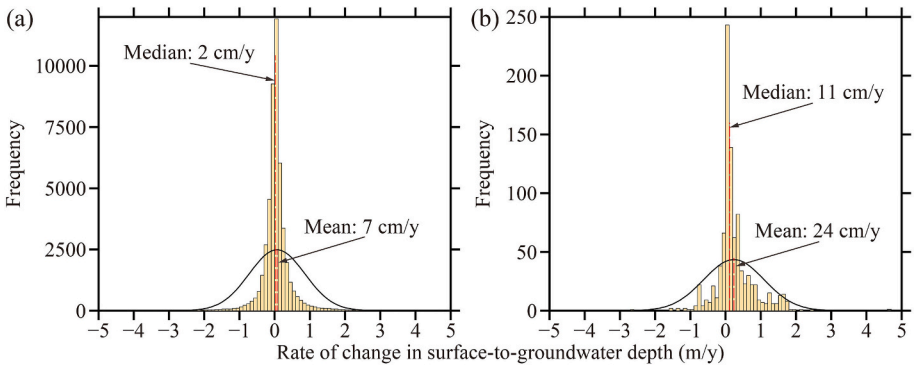

Fig. 9. Rate of change in surface-to-groundwater depth in LS and non-LS areas. (a) Rate of change observed in wells located outside LS areas. (b) Rate of change observed in wells located within LS areas. Groundwater level data sourced from Jasechko et al. (2024).

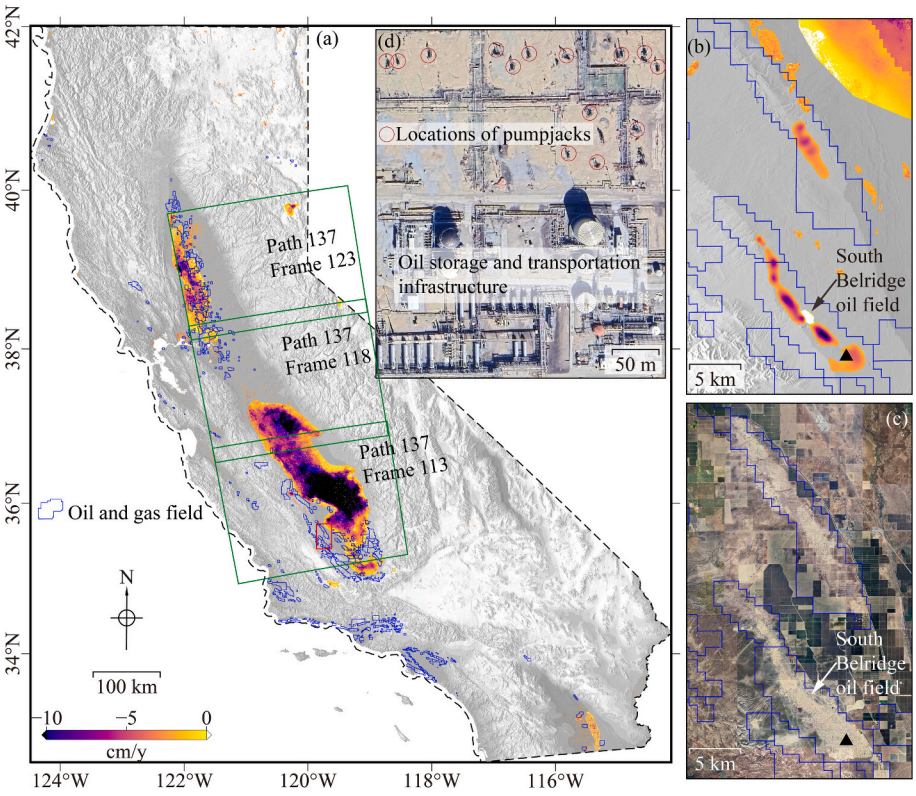

Fig. 10. Spatial relationship between oil and gas fields and LS in California. (a) Spatial distribution of oil and gas fields overlaid on the LS map. (b) Deformation rate within the South Belridge oil field. (c) Optical image of the South Belridge oil field. (d) Detailed view of pumpjacks and oil storage and transportation infrastructure in the South Belridge oil field. The location of (d) is marked by a black triangle in both (b) and (c).

challenging.

A key goal of SGMA is to reduce LS through groundwater regulation. However, InSAR observations indicate persistent subsidence in California's Central Valley, despite SGMA implementation (Fig. 7(a)). To assess temporal trends in subsidence after the implementation of SGMA, we analyzed GNSS data from the largest LS area in the CONUS (CA_LS1 in Fig. 7(a)). Analysis of 27 GNSS stations in CA_LS1 reveals diverse subsidence patterns: 4 stations show no subsidence, 16 exhibit continuous subsidence, and 11 show reduced subsidence with significant fluctuations (Figs. 7(a), S4, S5, and 11(a)). Although subsidence has nearly ceased at these 11 stations since 2023, this reduction cannot be attributed to the implementation of SGMA. The observed pattern—slower subsidence during wet years and accelerated subsidence during dry years—suggests that subsidence is strongly influenced by seasonal rainfall variability rather than by SGMA-related interventions. The cessation of subsidence following 2016 and 2022 corresponds with the conclusion of drought periods (2012–2016 and 2020–2022), indicating that subsidence levels could revert during future dry spells. Given that these stations spread across different groundwater basins, each with varying recharge timelines, it is unlikely that SGMA alone explains these changes. In addition, ongoing groundwater depletion (Liu et al., 2022a) also underscores the unintended consequences of SGMA and the need for further improvements in groundwater regulation.

The Santa Ana Basin, another area seriously impacted by groundwater extraction, has experienced long-term subsidence, with seasonal uplift, as well as a notable two-year period of uplift that begin in late 2004 (De Luca et al., 2017; Li et al., 2012; Riel et al., 2018). Our analysis of Sentinel-1 images from November 2015 to August 2024 shows a shift to long-term uplift, with seasonal fluctuations (Fig. 11(b)–(d)). Descending Sentinel-1 data from May 2015 to April 2018 also indicate uplift (Hou et al., 2021; Zhu et al., 2022). This trend, observed even during drought years, suggests that subsidence in the Santa Ana Basin has been effectively mitigated, primarily through long-term groundwater management, implementation of SGMA, and local water conservation efforts (Greg Woodside and Westropp, 2015; Riel et al., 2018). Effective subsidence management requires sustained efforts, with the primary objective being the regulation of groundwater. Methods to achieve this include restricting groundwater extraction, rejecting groundwater, and alleviating pressure on groundwater resources through the construction of reservoirs and canals.

The long-term management of water resources to mitigate LS is also demonstrated by other countries, such as China (Ao et al., 2024). In China, the government initiated the South-North Water Diversion Project, which transfers water from the south to the northern cities to alleviate pressure on groundwater resources. This project has partially mitigated LS in the Huabei Plain (Ao et al., 2024; Yu et al., 2023), although subsidence in the region has not ceased due to its vast extent and high subsidence rate. In contrast to the Central Valley, where subsidence remains severe, this highlights the need for sustained and integrated management strategies. Such contrast also underscores the challenges faced by regions experiencing severe subsidence. Compared to the Santa Ana Basin, the Central Valley faces greater challenges due to its higher subsidence rates, larger affected areas, and extensive groundwater-dependent agriculture, valued at $17 billion annually (Faunt, 2009). Addressing subsidence in the Central Valley requires a comprehensive approach that integrates groundwater regulation, sustainable agricultural practices, and long-term planning.

Climate changes exacerbate drought frequency in arid regions of the CONUS (IPCC, 2022), intensifying the need for effective LS management. California, for example, has experienced severe droughts from 2012 to 2016 and again from 2020 to 2022 (Liu et al., 2022a), which intensify the dependence on groundwater, complicating subsidence management. Farmers in drought-prone areas often rely on irrigation to sustain crops. In addition, shifting rainfall patterns further drive the

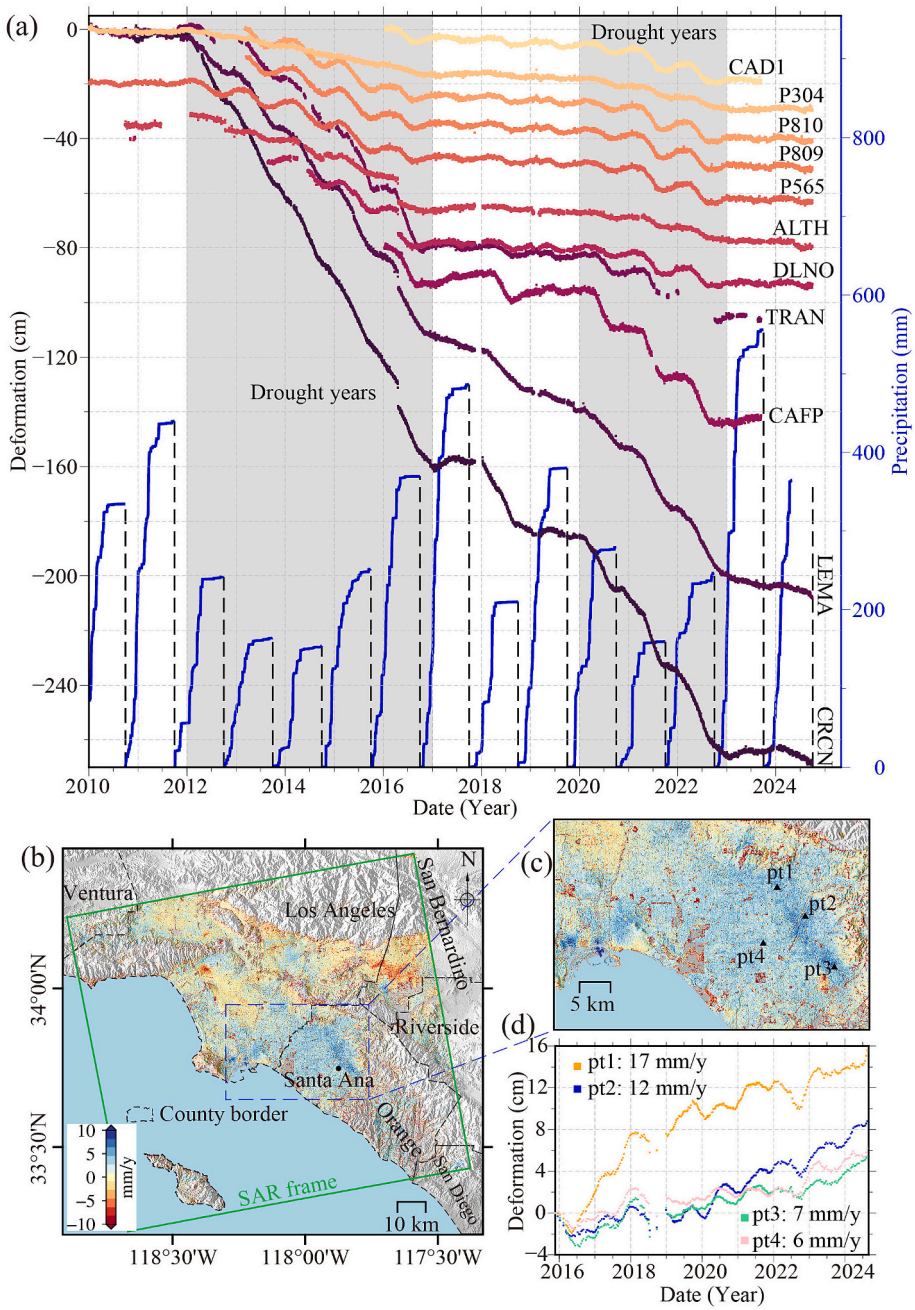

Fig. 11. Temporal deformation patterns in CA_LS1 and the Santa Ana Basin. (a) Vertical deformation time series from GNSS stations in CA_LS1, showing reduced subsidence rates. Dark green dots in Fig. 7(a) denote the locations of the 11 GNSS stations. The deformation time series for each station are shown in distinct colors. The cumulative precipitation, calculated over a water year (from October 1 of the previous year to September 30 of the current year), is overlaid for reference. (b) Average deformation rate across the SAR frame covering the Santa Ana Basin. (c) Enlarged view of the deformation rate within the purple-dotted rectangle in (b). (d) Deformation time series for points pt1, pt2, pt3, and pt4, as marked in (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

demand for groundwater extraction. In arid regions with extensive irrigated farmland, such as California, Arizona, Nebraska, Arkansas, Nevada, and Utah (Fig. 7), LS management is difficult. Effective LS management requires sustained effort, but few states have implemented stringent groundwater regulations. Local authorities must balance the need for long-term groundwater sustainability with short-term economic pressures, addressing both agricultural demands and environmental sustainability.

4.3. Potential risks caused by LS

LS in the US presents a cascade of risks that extend beyond immediate infrastructure damage and urban development challenges. In the western and southern US, excessive groundwater extraction has led to significant subsidence, particularly affecting agricultural regions. This subsidence alters land elevation, disrupting hydrological processes, reducing groundwater storage capacity (Hasan et al., 2023; Hilton and Jasechko, 2023; Jasechko et al., 2024), and increasing floodplain vulnerability to extreme weather events. Such hydrological imbalances contribute to longer-term challenges, including heightened flood risks in

coastal and riverine zones. The combined effect of subsidence and rising sea levels accelerates coastal inundation, particularly in the San Francisco Bay Area (Blackwell et al., 2020), as well as along the Gulf Coast and East Coast (Ohenhen et al., 2023b; Ohenhen et al., 2024), exacerbating infrastructure strain, including stormwater drainage systems. This dynamic leads to costly repairs and increased public safety risks. Given these regions' heavy reliance on water resources for agriculture and urban supply, subsidence represents a major socio-economic threat, further complicating efforts to achieve sustainable development goals, particularly within the framework outlined by the United Nations (Nations, 2015).

Subsidence in wetland areas along the East Coast and Gulf Coast has caused profound ecological and environmental consequences. Wetland subsidence, driven by both natural processes and human activities (Jankowski et al., 2017), diminishes the wetlands' capacity to function as carbon sinks (Xu et al., 2024). As these areas subside, stored carbon is released into the atmosphere, further exacerbating climate change. Additionally, wetland subsidence increases the risk of saltwater intrusion into coastal aquifers, threatening local freshwater supplies. This, coupled with the loss of habitats for migratory species and biodiversity, intensifies environmental degradation (Thorne et al., 2018). Effective management of wetland sink requires an integrated approach that combines environmental conservation, urban planning, and climate adaptation strategies.

The findings presented in this study, including a national LS map for the CONUS, provide essential data to support localized subsidence management. However, the implications extend far beyond the CONUS. Globally, subsidence poses similar risks, particularly in coastal and riverine regions already vulnerable to the effects of climate change and sea-level rise. Given the global nature of LS (Hasan et al., 2023; Herrera-García et al., 2021), the methodologies and findings serve as a crucial reference for other countries and regions, offering a foundation for international efforts to mitigate the impacts of subsidence in the context of global climate change, sustainable development, and rising sea levels.

4.4. Effectiveness of methodology and results comparison

4.4.1. Effectiveness of the methods used

In this section, we demonstrate the effectiveness of the methods employed, including EDAD correction, mosaicking of deformation results, and identification of deformation areas, through the following three subsections.

4.4.1.1. EDAD correction. We use the initial average deformation rates (Fig. 12), which were obtained from different processing methods without an iterative process, to demonstrate the effectiveness of the proposed EDAD correction method. For simplicity, the four methods illustrated in Fig. 12(a)–(d) as method1 to method 4, respectively. Method 1 only removes the phase ramp. Method 2 uses global fitting, without multi-patch, to correct EDAD. Method 3 employs the multi-patch method to estimate and correct EDAD within each patch, with patch size and overlap consist with those used in the proposed method. Method 4 is the proposed method as detailed in Section 2.2.1. The processed area (from Path 137, Frame 123, coverage of this path is shown in Fig. 10(a)) covers both plain and mountainous regions. Notable large-scale LS signals are observed in several regions, such as area A1 and A2, both located on cultivated lands (see "identification of deformation areas" section). In particular, area A1, situated in the Central Valley of California, is a recognized zone of significant subsidence.

Method 3, which relies on multi-patch fitting, effectively corrects EDAD but is suitable only in mountainous areas, as it removes most of the LS signals (Fig. 12(c)). In contrast, methods 1, 2 and 4 preserve LS signals. However, methods1 and 2 retain obvious EDADs in the deformation results, especially within the mountainous area marked by the dotted ellipse. The proposed method (method 4) generates a deformation rate field that effectively mitigates EDAD while preserving true deformation signals. By masking out significant deformation areas, such as areas A1 and A2, the SDs of the deformation rates obtained from

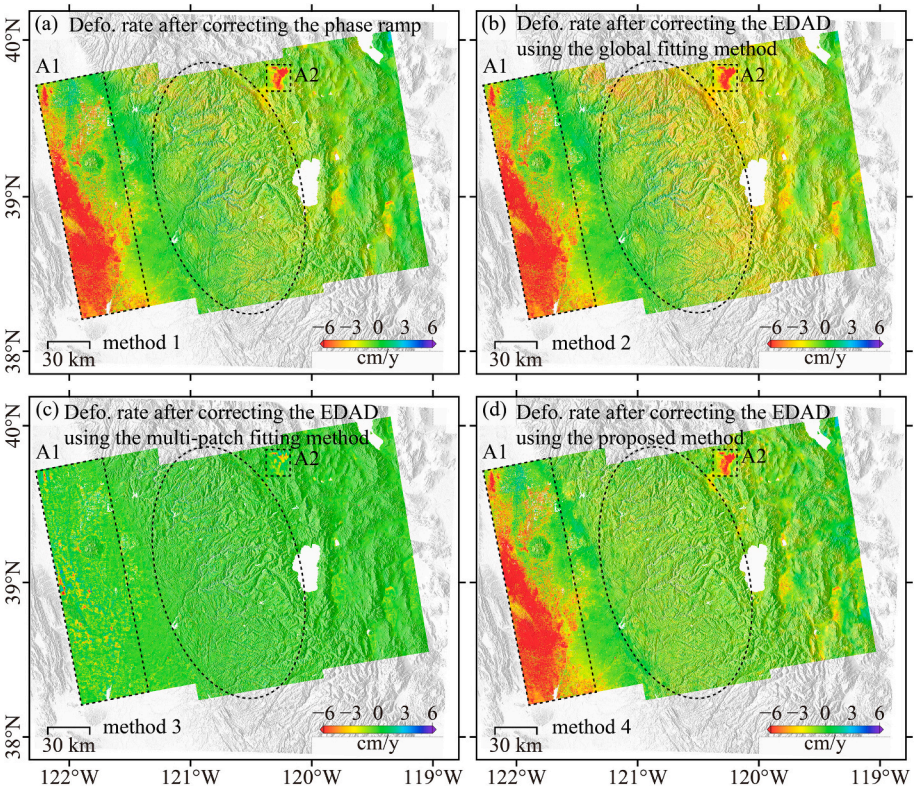

Fig. 12. Average deformation rates derived from different processing methods for Path 137, Frame 123: (a) method 1, (b) method 2, (c) method 3, and (d) method 4.

method 1, method 2, and method 4 are 11 mm/y, 12 mm/y, and 8 mm/y, respectively, further demonstrating the reliability of method 4. Therefore, the proposed method can provide a reliable initial estimate for subsequent iterative processing.

4.4.1.2. Mosaicking of deformation results. Fig. 2 illustrates that the deformation rate map after mosaicking exhibits greater consistency than the pre-mosaicking result. We employ the average deformation rates to quantitatively evaluate the consistence of the used mosaicking method. Since the derived deformation rates are referenced to local points within SAR images, and given that most SAR images, except those capturing large-scale signals such as substantial LS or coseismic deformation, contain only minor deformation components, the average deformation rates for most SAR frames are expected to be close to zero. However, biases may arise due to reference points or errors. The mosaicking process can mitigate such systematic biases, thereby improving the overall consistency of deformation results. This theoretical rationale supports the use of average deformation rates as a metric for assessing the effectiveness of the mosaicking method.

As shown in Fig. 13, the average deformation rates after mosaicking are more clustered around zero compared to those before mosaicking. Nonetheless, certain frames, such as Path 137, Frames 113 and 118, still exhibit notable deviations from zero even after mosaicking, which can be attributed to their coverage of extensive LS areas. Specifically, these frames encompass California's Central Valley (see Fig. 10(a)), a region experiencing the most severe LS in terms of both rate and spatial extent (Fig. 7(a)). The SD and Mean Average Deviation (MAD) relative to zero for the average deformation rates before mosaicking were 2.9 mm/y and 2.1 mm/y, respectively. After mosaicking, there values were significantly reduced to 1.4 mm/y and 0.7 mm/y. This marked improvement in both statistical measures proves the effectiveness of the mosaicking method.

4.4.1.3. Identification of deformation areas. Our deep learning network, DeforNet, was trained on an extensive dataset of InSAR deformation samples acquired from various regions in China (details in Section 2.2.3). Given that our model leverages the deformation rate fringes to identify subsidence, and the wrapped deformation rates exhibit high similarity across different areas, we anticipate that DeforNet can generalize effectively to other regions beyond its training domain. To evaluate its accuracy, we selected a test site within the coverage of Path 137, Frame 123. This area is characterized by widespread LS (Fig. 12), diverse topography including plains and mountains, and significant atmospheric delays. These challenging conditions make it an appropriate validation site for assessing the accuracy of our LS identification model.

We identified LS areas using the iteratively processed deformation rate (see Section 2.2 for details). A total of 701 potential LS areas were detected (Fig. 14), the majority of which were located in cultivated land (Fig. 14(b)). These areas were visually interpreted using Google Earth™, among which 91 were classified as false LS areas. False LS identifications were determined based on the following criteria: 1) the area was not situated in typical LS-prone land covers such as cultivated land, wetlands, construction zones, or oil and gas fields; and 2) the deformation pattern was inconsistent the surrounding areas. It should be noted that false LS areas do not necessarily indicate the complete absence of subsidence, but rather reflect uncertainties in classification. Based on this validation, the overall accuracy of LS identification was about 87 %.

The impact of false LS areas on subsidence area statistics is limited, as

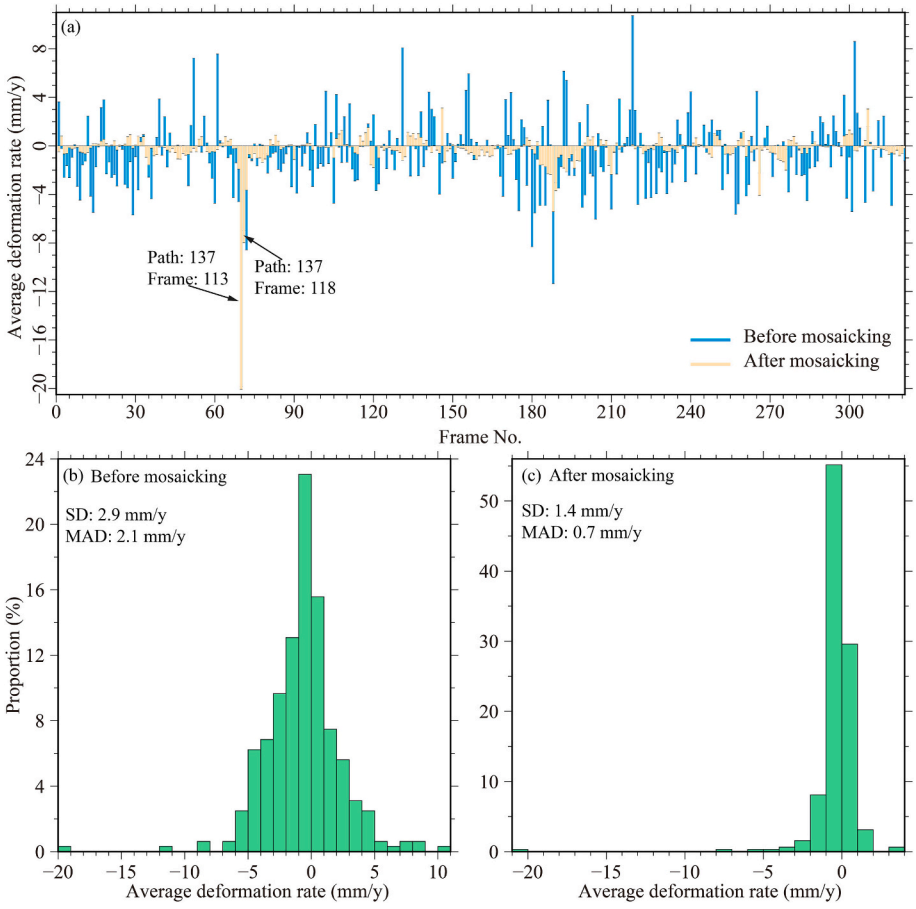

Fig. 13. Comparison of average deformation rates before and after mosaicking. (a) Average deformation rates per frame before and after mosaicking. (b) Distribution histograms of the average deformation rates before and after mosaicking.

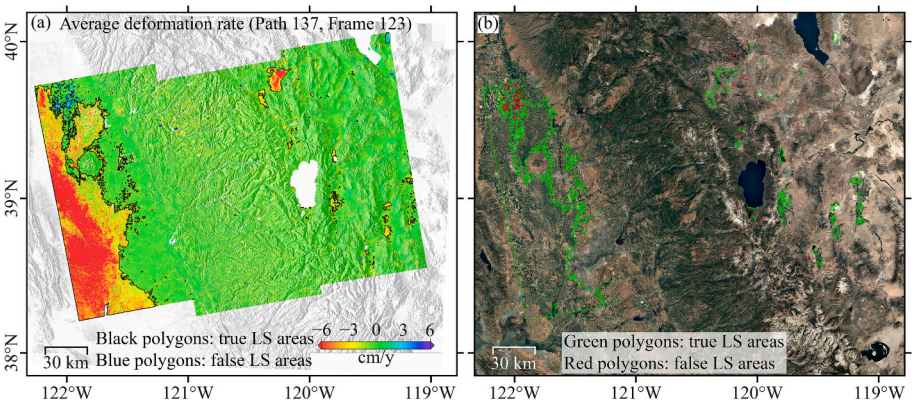

Fig. 14. Average deformation rate and corresponding LS identification results. (a) Deformation rate map (Path 137, Frame 123) from the iterative processing, alongside the LS identification results. (b) Optical image of the same scene with superimposed LS identification results.

these areas constitute only a small proportion of the total identified LS regions. This is largely because false LS areas typically originated from minor errors in the deformation results, which are negligible compared to the spatial extent of true subsiding areas. For example, in the test site, the false LS areas account for only 1 % of the total LS-related pixels. Therefore, the accuracy of our model is sufficient for reliable identification and statistical analysis of LS.

4.4.2. *Validation and comparative analysis of results*

Validating all identified LS areas at a national scale is challenging. In addition to GNSS verification (Section 3.1), we conduct validation and comparative analysis of the deformation results using published data. Table 2 provides a detailed description of the validation process for the major subsidence areas that coincide temporally with our study. Table 2 shows that the LS areas identified in this study are basically consistent with the public data, both in terms of spatial extent and subsidence rates.

Some studies have also assessed the population and areas exposed to LS in the US at a regional scale (Ohenhen et al., 2023a) or in major cities (Ohenhen et al., 2025). The estimates of exposed population and areas reported in these studies are larger than those obtained in our study, even though our analysis covers the entire CONUS. This discrepancy primarily arises from differences in the time period of SAR data used (Ohenhen et al., 2023a: 2007–2020; Ohenhen et al., 2025: 2015–2021), reference frames, and LS identification methods.

Table 2
Validation of the deformation results against published data.

Validation sites	Time period	Sources	Spatial consistency	Rate consistency
Fort Grant Rd and Willcox areas in Arizona	Apr. 2019–Apr. 2020	Arizona department of water resources. SAR data: Radarsat-2	Aligns with area AZ_LS1 in Fig. 7(c).	Comparable
Willcox and Kansas settlement areas	Apr. 2019–Apr. 2020	Arizona department of water resources. SAR data: Radarsat-2	Aligns with areas AZ_LS2 and AZ_LS3 in Fig. 7(c).	Comparable
Elfrida area	Apr. 2019–Apr. 2020	Arizona department of water resources. SAR data: Radarsat-2	Aligns with areas AZ_LS2 and AZ_LS4 in Fig. 7(c).	Comparable
Bowie and San Simon areas	Mar. 2021–Jan. 2022	Arizona department of water resources. SAR data: Sentinel-1	Aligns with areas AZ_LS5 and AZ_LS6 in Fig. 7(c).	Comparable
Arizona	Partially coincides with the period of our study.	Arizona department of water resources. SAR data: Radarsat-2 / Sentinle-1	Aligns with the deformation areas obtained by this study. The detailed deformation rate map was uploaded on our website: at http://47.119.124.134/insarclub/, accessible through a free account registration.	Comparable
Santa Ana Basin in California	May 2015–Apr. 2018	Hou et al., 2021; Zhu et al., 2022 SAR data: descending Sentinel-1	Aligns with the deformation areas obtained by this study (Fig. 11(b)).	Our deformation rate is slightly lower than their results. The temporal coverage of the Sentinel-1 data used in this study spans from Nov. 2015 to Aug. 2024.
Central Valley in California	Aug. 2019–Sep. 2021	Carlson et al., 2024 SAR data: descending Sentinel-1	Aligns with the deformation areas obtained by this study (Fig. 10).	Comparable

Our deformation results were referenced to local reference points, whereas their deformation data were tied to IGS14 global reference frame. In their approach, LS pixels were defined as those exhibiting a Vertical Land Motion (VLM) rate less than 0 mm/y. In contrast, we employed a deep learning-based method that accounts for the spatial clustering of deformation pixels. As a result, many pixels that did not exhibit cluster-like behavior consistent with LS were not classified as such in our analysis, due to potential uncertainties in the deformation results. This is further supported by the RMSE of the deformation rates, which is 5.2 mm/y when validated against GNSS observations. Additionally, the choice of reference frame can significantly influence LS identification. In a local reference frame, trend signals are removed, whereas a global reference frame retains these signals. This difference in signal processing leads to variations in LS delineation, ultimately affecting estimates of subsidence area and exposed population. In addition to the area of LS and population, the aforementioned differences also influence the assessment of risk levels, which are typically evaluated based on both angular distortions (correlated with temporal coverage and deformation rates) and the population/houses exposed to LS. Our study uses the same classification of angular distortions as the studies referenced. However, the classification of exposed houses/population varies depending on the specific application (Text S4).

The abovementioned methods represent two main categories of LS identification: 1) using deformation rates to extract LS pixels (e.g., Ohenhen et al., 2024; Ohenhen et al., 2025); and 2) defining LS areas based on the clustering features of deformation pixels (e.g., Costantini et al., 2017; Herrera-García et al., 2021; Qiao et al., 2023; Zhang et al., 2024). The first category usually references deformation results to a global reference frame, which preserves long-wavelength trend signals. These two methods are complementary, and the selection between them should be guided by the specific application.

5. Conclusion

Leveraging over twenty thousand Sentinel-1 images, we quantified and analyzed LS across the CONUS, providing unprecedented insights into its spatial extent, potential influencing factors, and socio-economic impacts. This work lays a critical foundation for the development of targeted management and mitigation strategies. The main conclusions are summarized as follows:

(1) LS occurs in all the 48 contiguous states and the District of Columbia, covering a total area of approximately 45,666 km^2. California experiences the most severe subsidence, both in terms of magnitude and affected area, with the maximum LOS subsidence reaching up to 47.5 cm/y.
(2) Approximately 73 % of subsidence occurs in cultivated lands, and 10 % in wetlands, with groundwater overexploitation emerging as the primary driver. This analysis highlights the critical relationship between agricultural practices, water resource management, and subsidence.
(3) Over 6 million people, 160,000 buildings, and thousands of kilometers railways and roads are located in subsidence-affected areas.

By evaluating temporal changes in subsidence across key regions, we further demonstrate the challenges of mitigating subsidence caused by groundwater extraction. Our analysis highlights the need for a multifaceted approach that integrates groundwater regulation, sustainable agricultural practices, and long-term planning. These findings have profound implications for water resource management, infrastructure resilience, and climate adaptation strategies, particularly in the context of increasing drought frequency and intensity.

This study has several limitations regarding the capture of long-wavelength deformation signals and the analysis of the temporal evolution of LS. Since the deformation results were referenced to local points, long-wavelength signals, such as those induced by tectonic activity, glacial isostatic adjustment, and other regional factors, were not preserved. Furthermore, deformation time series were only computed for specific areas, limiting the insight into the dynamics of LS. For future work, targeted regions could be selected based on the deformation rate map for more detailed investigation. Follow-up studies may involve geotechnical and geophysical modeling to better understand the driving mechanisms behind LS.

CRediT authorship contribution statement

Zhiqiang Xiong: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Guangcai Feng:** Writing – review & editing, Funding acquisition, Data curation, Conceptualization. **Zefa Yang:** Writing – review & editing, Validation, Data curation. **Hua Gao:** Writing – review & editing, Validation, Data curation. **Wenbin Xu:** Writing – review & editing, Validation, Data curation. **Lei Zhang:** Writing – review & editing, Validation. **Xiaohua Xu:** Writing – review & editing. **Zhong Lu:** Writing – review & editing. **Jun Hu:** Data curation. **Zhiwei Li:** Writing – review & editing. **Jianjun Zhu:** Writing – review & editing.

Funding

This research was supported by the National Natural Science Foundation of China (Grant No. 42174039).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors express their gratitude to the Alaska Satellite Facility for providing On-Demand InSAR services. The authors acknowledge the GMT open-source software. We acknowledge the providers of all open-source datasets and research articles. We thank the graduates in our research group for downloading the InSAR data. The authors would like to express their gratitude to the three anonymous reviewers for their valuable comments and suggestions.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2025.115134.

Data availability

The Sentinel-1 images and interferograms used in this study are publicly available and can be downloaded from Alaska Satellite Facility at https://search.asf.alaska.edu/. Data on roads and buildings were provided by OpenStreetMap and are publicly available for download at https://www.openstreetmap.org/. The descriptions and sources of the auxiliary datasets, including land cover, rainfed and irrigated crops, population data, precipitation data, GNSS, aquifers and groundwater data, used in the analysis of land subsidence are provided in the Supplementary Information. The codes used to process InSAR data are available upon reasonable request. The deformation maps of the CONUS and other countries are available on our website at http://47.119.124.134/insarclub/, accessible through a free account registration.

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